



MASTERSCAN ENGINEERING

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GST/Co. Reg. No.: 200506867D 29 Benoi Road, Pioneer Lot, Singapore 629922



REPORT NO : MSE/JEL/CCP4/MPI/F041

MSE/REF/NO : D2-26-2777

TESTED DATE : 17.04.2026

PAGE : 1 OF 4

MAGNETIC PARTICLE TEST REPORT

CLIENT : JURONG ENGINEERING LIMITED
OWNER : SEMBCORP COGEN PTE LTD
CONTRACTOR : SUNVIEW ENGINEERING PTE LTD
LOCATION : BANYAN CCP4 SITE
PROJECT TITLE : BANYAN CCP4 PROJECT
REQUEST NO : REQ-CCP4-MT-F013
DESCRIPTION : MT for Carbon Steel
ITEM TESTED : WELDED JOINTS
TEST PROCEDURE / DOC NO : CCP4-SCC3-BY20-Q13-4-033_1
ACCEPTANCE SPEC : ASME B31.1 : 2022 Ed (Para. 136.4.3)

TEST METHOD : VISIBLE WET CONTINUOUS METHOD
EQUIPMENT : CONTOUR PROBE, AC ELECTROMAGNETIC YOKE (SR.NO : 14046)
POLE SPACING : 3 INCHES TO 6 INCHES
CALIBRATION : FIELD INDICATIONS STRIP AND 10 LBS LIFTING BLOCK(MSE-LB-03)
SURFACE CONDITION : AS WELDED & CLEANED
MATERIAL : A106 Gr.C
SURFACE TEMPERATURE : TEMPERATURE (25 TO 35°C)
CONTRAST PAINT : MR72 WHITE PAINT (LOT NO :2401059)
MAGNETIC INK : MR76S BLACK INK (LOT NO : 2401048)
PARTICLE APPLICATION : SPRAYING
LIGHT METER SERIAL NO : P20008686
LIGHT/SOURCE INTENSITY : NATURAL LIGHT > 1080 Lux

SCOPE OF EXAMINATION : MAGNETIC PARTICLE TESTING WAS CARRIED OUT ON WELDED JOINTS.
AREA OF EXAMINATION : TESTED LOCATION PLEASE REFER TO THE ATTACHED DATA & DRAWING SHEETS.
FINDINGS : NO RELEVANT INDICATIONS WERE DETECTED AT THE TIME OF TESTING.
ITEM TESTED WERE FOUND COMPLIED TO SPECIFICATION.
RESULT : ACCEPTABLE

Tested and Reported By :
A.VENKATESWARA
NDT LEVEL II



DATE : 17.04.2026

Reviewed by:
SURENDRAKUMAR P.V
ASNT NDT Level III



DATE : 18.04.2026

Reviewed by:
JEL

DATE :

Reviewed by:
SEMBCORP COGEN

DATE :

FORM NO : MSE/NDT/MT/001 Rev.2

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PAGE : 2 OF 4

S/N	ISO DRAWING	LINE NO	REV	JOINT NO	SIZE-THK	MATERIAL	WELDER NO		WPS NO	SHT NO	WELDING PROCESS	WELD TYPE	Distance from Reference Point	Type of Indication	RESULT	REMARKS
							ROOT	COVER								
1	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR102	-	4A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	29	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
2	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR108	-	20A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	12	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
3	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR108	-	16A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	12	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
4	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR108	-	4A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	12	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
5	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR108	-	2A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	12	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
6	CCP4-SCC3-BY21-P12-3-287	68LAB92-BR142	-	8A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	25	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
7	CCP4-SCC3-BY21-P12-3-287	68LAB92-BR142	-	12A	PL10x40x100	A106 Gr.C	PW-053	PW-053	JWPS-547	25	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
8	CCP4-SCC3-BY21-P12-3-287	68LAB83-AA912	-	1	1"x4.55	A106 Gr.C	PW-202	PW-202	JWPS-547	50	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
9	CCP4-SCC3-BY21-P12-3-287	68LAB83-AA912	-	2	1"x4.55	A106 Gr.C	PW-202	PW-202	JWPS-547	50	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
10	CCP4-SCC3-BY21-P12-3-287	68LAB83-AA912	-	3	1"x4.55	A106 Gr.C	PW-202	PW-202	JWPS-547	50	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
11	CCP4-SCC3-BY21-P12-3-287	68LAB83-AA912	-	4	1"x4.55	A106 Gr.C	PW-202	PW-202	JWPS-547	50	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
12	CCP4-SCC3-BY21-P12-3-287	68LAB83-AA912	-	6	1"x4.55	A106 Gr.C	PW-186	PW-186	JWPS-547	50	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
13	CCP4-SCC3-BY21-P12-3-287	68LAB83-AA912	-	6	1"x4.55	A106 Gr.C	PW-186	PW-186	JWPS-547	50	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
14	CCP4-SCC3-BY21-P12-3-287	68LAB83-AA912	-	7	1"x4.55	A106 Gr.C	PW-186	PW-186	JWPS-547	50	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
15	CCP4-SCC3-BY21-P12-3-287	68LAB93-BR051	-	001	1.1/2"x10.15	A106 Gr.C	PW-200	PW-200	JWPS-547	36	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER

Tested and Reported By :
A.VENKATESWARA
NDT LEVEL II



DATE : 17.04.2026

Reviewed by:
SURENDRAKUMAR P.V
ASNT NDT Level III



DATE : 18.04.2026

Reviewed by:
JEL

DATE :

Reviewed by:
SEMBCORP COGEN

DATE :

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PAGE : 3 OF 4

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							ROOT	COVER								
16	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR801		001	1.1/2"x5.08	A106 Gr.C	PW-200	PW-200	JWPS-547	37	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
17	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR803		001	1.1/2"x5.08	A106 Gr.C	PW-053	PW-053	JWPS-547	38	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
18	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR803		002	1.1/2"x5.08	A106 Gr.C	PW-053	PW-053	JWPS-547	38	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
19	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR841		001	1.1/2"x10.15	A106 Gr.C	PW-187	PW-187	JWPS-547	39	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
20	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR841		002	1.1/2"x7.14	A106 Gr.C	PW-187	PW-187	JWPS-547	39	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
21	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR111		7A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	15	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
22	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR111		17A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	15	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER
23	CCP4-SCC3-BY21-P12-3-287	68LAB83-BR111		18A	PL10x40x100	A106 Gr.C	PW-187	PW-187	JWPS-547	15	GTAW	SW	-	-	ACCEPTABLE	FGH FEED WATER

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NDT LEVEL II



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PAGE : 4 OF 4



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